

AI EPC Platform - Engineering Specification

Engineering Specification

Project : Alpha Refinery

Vendor : Honeywell

Equipment : Boiler

Issue Date : 2026-03-02

Project Overview

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The EPC contractor shall deliver a fully operational facility including civil, mechanical, electrical and commissioning works.

The project consists of the construction of a Tier III data center supporting mission-critical operations.

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Cooling System

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Supply air temperature shall remain between 18°C and 27°C.

Each cooling unit shall include vibration monitoring sensors.

Chilled water pumps shall operate with redundant motors.

CRAH units shall support automatic failover.

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Cooling equipment shall support SNMP monitoring.

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The cooling system shall provide N+1 redundancy.

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Electrical System

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Generator backup shall support 72 hours of continuous operation.

UPS capacity shall be 500 KVA.

Electrical panels shall comply with IEC standards.

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Power Distribution Units shall provide branch circuit monitoring.

Emergency shutdown shall disconnect non-critical loads.

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Dual power feeds shall be provided for critical equipment.

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Fire Protection

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VESDA detection shall be installed.

Emergency evacuation procedures shall be displayed.

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Smoke detectors shall be installed in all electrical rooms.

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Fire alarm panels shall integrate with the BMS.

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FM200 suppression shall protect all server rooms.

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Network Infrastructure

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All network racks shall use Cat6A cabling.

Fiber optic backbone shall use single-mode fiber.

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Core switches shall support redundant uplinks.

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Dual fiber entry points shall be provided.

Network monitoring shall integrate with the NOC.

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Commissioning

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Integrated System Testing shall be completed before handover.

All UPS systems shall pass load testing.

Generator synchronization tests shall be completed.

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Quality Assurance

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All materials shall comply with project specifications.

Vendor documentation shall be reviewed before installation.

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All construction activities shall follow QA/QC procedures.

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